LIBBY — BIOMARKER SURVEY

urothelial-mets-her2-discordant-kndl

Selected general biomarker report — biomarker screening coverage and gaps

Generated 2026-08-20

Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

Surveyed: 23 biomarkers. **No usable result on file:** 11. **Measured but not decision-grade:** 9. **Already established:** 3.

The survey covered 23 biomarkers: all 9 tumor-agnostic markers plus 14 protein targets in scope for urothelial carcinoma. Three have hardened answers. The tumor is microsatellite stable, MAGE-A4 is negative, and HLA-A*02:01 is confirmed positive. The other 20 have no result a trial or a label could act on today. One gap is essential: HER2 status on the current tumor. Three validated IHC 3+ results exist, but they describe 2023 and 2025 tissue, and the only read of the 2026 nodal disease is a weak research-grade signal. A validated repeat IHC was drawn in 2026-08; its tier is missing from the record, not from the lab.

Closing the list takes three orders, none of which is a biopsy. First, a records retrieval of sequencing already performed, which reads out NTRK, RET, BRAF, and somatic HR status, recomputes TMB to companion-diagnostic grade, and names the one OncoKB-actionable alteration the record counts but never identifies. Second, retrieval of the 2026 assay reads: the validated HER2 and TROP2 IHC tiers, plus the Nectin-4, PRAME, and FAP imaging results. Third, one archival block, most plausibly the 2025-03 recurrence, cut once for HER2 with reflex ISH, TROP2, mesothelin, MUC1, and B7-H3, with PRAME and NY-ESO-1 run on the same block. That fits the tissue reality here: nodal cores carry 5-10% tumor content and one fresh-frozen vial remains.

Most of these, if tested, will come back negative. That is the expected result for rare markers in this disease. The case for testing is what a positive would open, cell-therapy and ADC trial classes among them, not an implied likelihood of finding one. The weak TROP2 signal stays a low positive, and a low positive predicts ADC benefit less reliably than strong expression does.

Gaps: biomarkers with no usable result on file

BRAF V600E mutation

Priority: medium **Scope:** Tumor-agnostic **Status:** Record ambiguous

BRAF V600E is actionable across primary sites outside colorectal cancer. A 468-gene panel of the class used here covers BRAF codon 600, but no BRAF result, positive or negative, is in the derived record.

BRAF V600E is uncommon in urothelial carcinoma and the honest expectation is that it is absent. It stays on this list because the record explicitly counts one OncoKB-actionable alteration and never says which, so the identity of that variant is simply unknown. Reading it off the existing report costs a records request.

Recommended assay. Read BRAF codon 600 status off the retrieved full tissue NGS report (NGS_panel · None; the sequencing is already done · days)

Bundles with. Retrieval of the sequencing already performed (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), which also returns FGFR3 status and the single unnamed OncoKB-actionable alteration. No new specimen needed

A positive result would open.

- dabrafenib plus trametinib (BRAF inhibitor plus MEK inhibitor · tumor-agnostic approval · NCT02034110)

Next step. Bundle with planned testing

References. nct:NCT02034110

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Record ambiguous

Any solid tumor with an in-frame NTRK fusion qualifies for TRK inhibition, primary site aside. Fusion-capable sequencing has been performed here, but no NTRK result appears in the derived record.

This is a records problem, not a testing problem: whole-transcriptome sequencing that would show an NTRK fusion has already been run on two platforms and the reports are not in hand. NTRK fusions in urothelial carcinoma are rare, so a negative is the expected answer, and the reason to settle it is that the same retrieval also names the single OncoKB-actionable alteration the record counts but never identifies. A DNA-only panel reporting no structural variants does not close the question, because NTRK breakpoints sit in large introns.

Recommended assay. Retrieve the whole-transcriptome and NGS reports and read the NTRK fusion status directly; RNA fusion panel only if none covered NTRK (RNA_seq · None for retrieval; archival FFPE if an RNA fusion panel turns out to be needed · days for retrieval, 2-3 weeks for a new RNA panel)

Bundles with. Retrieval of the sequencing already performed (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), which also returns FGFR3 status and the single unnamed OncoKB-actionable alteration. No new specimen needed

A positive result would open.

- larotrectinib (TRK inhibitor · tumor-agnostic approval · NCT02576431)
- entrectinib (TRK / ROS1 / ALK inhibitor · tumor-agnostic approval · NCT02568267)
- repotrectinib (next-generation TRK inhibitor · tumor-agnostic approval · NCT03093116)

Next step. Bundle with planned testing

References. nct:NCT02576431; nct:NCT02568267; nct:NCT03093116

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Record ambiguous

Selpercatinib is approved for RET fusion-positive solid tumors irrespective of site. As with NTRK, the sequencing that would answer this has been done and the reports have not been retrieved.

RET fusions are rare in urothelial carcinoma and nothing in this record suggests one, so the expected result is negative. It rides free on the report retrieval that this case needs anyway, and leaving it unread means an approved oral agent stays formally unassessed. A DNA panel without RET intronic coverage would not be sufficient on its own.

Recommended assay. Read RET fusion status off the retrieved transcriptome or NGS report; RNA fusion panel only if RET was not covered (RNA_seq · None for retrieval · days for retrieval)

Bundles with. Retrieval of the sequencing already performed (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), which also returns FGFR3 status and the single unnamed OncoKB-actionable alteration. No new specimen needed

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128

B7-H3 (CD276) protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

B7-H3 is overexpressed across most solid tumor types with limited normal surface expression, and B7-H3-directed ADCs and bispecifics are enrolling solid-tumor cohorts. Nothing in this record speaks to it.

B7-H3 has never been looked at in this patient, and it is the pan-cancer surface target with the deepest current trial bench. One extra slide from the block already being cut for HER2 and TROP2 answers it, which is the only reason it earns a place against a tissue-scarce case. Expression would open solid-tumor ADC cohorts; the payload class would still have to be weighed against the pulmonary findings.

Recommended assay. B7-H3 (CD276) IHC on tumor tissue (IHC · Archival block, one additional slide from the cut already planned · 1 week)

Bundles with. One archival block cut for the validated IHC set: HER2 with reflex ISH, TROP2, mesothelin, MUC1 and B7-H3 read together on the 2025-03 recurrence or the 2026-08 nodal core

A positive result would open.

- ifinatamab deruxtecan (B7-H3-directed ADC · trial only)
- enoblituzumab (B7-H3-directed mAb · trial only)
- vobramitamab duocarmazine (B7-H3-directed ADC · trial only)

Next step. Bundle with planned testing

NY-ESO-1 (CTAG1B) expression, HLA-A*02:01-restricted

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

NY-ESO-1 is the other HLA-A*02:01-restricted cancer/testis antigen with a clinical-stage TCR-T program, and this patient's HLA gate is already satisfied. Expression has never been tested.

The patient is HLA-A*02:01 positive with an active adoptive-cell-therapy programme, and NY-ESO-1 is the second A*02:01-restricted antigen with an open TCR-T bench. Expression in urothelial carcinoma is a minority finding, so a negative is the likely outcome, and it costs one assay run next to the PRAME order. MAGE-A4 already showed how quickly an expression assay closes one of these axes.

Recommended assay. NY-ESO-1 expression by IHC or RT-PCR, run alongside PRAME on the same block (IHC · Archival block; RT-PCR is the more forgiving option at 5-10% tumor content · 1-2 weeks)

Bundles with. Cancer/testis antigen expression panel (PRAME, NY-ESO-1) by validated IHC or RT-PCR on the same archival block, alongside the retrieval of the 2026-08 PRAME imaging read

A positive result would open.

- letetresgene autoleucel (NY-ESO-1-directed TCR-T · trial only)

Next step. Bundle with planned testing

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Record ambiguous

PARP inhibitors are approved in other tumor types and not in urothelial carcinoma, but DNA-damage response alterations gate trial entry, and this patient already has an ARID1A-driven DDR hypothesis in play. Germline testing was negative; somatic HR status is not in the record.

Germline testing came back negative, which leaves somatic HR gene status as the unanswered half, and it sits in a report that has not been retrieved. Urothelial carcinoma is not a PARP-inhibitor indication, so the realistic payoff is trial eligibility and not an approved drug, and it lands next to the ARID1A synthetic-lethality hypothesis already on the feature list. No new specimen is involved.

Recommended assay. Read somatic HR gene status off the retrieved tissue NGS report; add an HRD score only if a protocol requires one (NGS_panel · None for retrieval · days for retrieval)

Bundles with. Retrieval of the sequencing already performed (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), which also returns FGFR3 status and the single unnamed OncoKB-actionable alteration. No new specimen needed

A positive result would open.

- PARP inhibitors (PARP inhibitor · approved, other tumor type)

Next step. Bundle with planned testing

EPHB4 protein expression

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

EPHB4 is overexpressed on bladder and prostate cancers with a tumor-vascular component, which puts it in scope by tumor type. Its clinical bench is one fusion protein that has not advanced.

EPHB4 earns a line on a bladder gap list and nothing more: the evidence tier is preclinical, the one clinical-stage binder was studied in other diseases, and endothelial expression makes the therapeutic window uncertain. Spending scarce tissue on it would displace stains with real options behind them.

Recommended assay. EPHB4 IHC on tumor tissue (IHC · Archival block · 2 weeks, research assay)

A positive result would open.

- soluble EphB4-HSA (fusion protein · trial only)

Next step. Defer

Integrin alpha-3 (ITGA3) protein expression

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

ITGA3 is upregulated on bladder carcinoma and drives invasion, so the panel puts it in scope for this tumor type. There is no anti-ITGA3 binder in clinical development.

ITGA3 has no clinical-stage binder, so a result either way would change nothing that can be acted on today. The row exists so the reader can see the target was considered and set aside, not because a stain is worth ordering. Broad normal epithelial expression would also limit any future window.

Next step. Defer

PSCA protein expression

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

PSCA is upregulated in bladder cancer as well as in prostate and pancreatic cancer, and this patient also carries a treated prostate primary. No PSCA result exists.

PSCA is plausibly expressed in this tumor but the clinical bench is confined to early prostate and pancreatic CAR-T protocols, none of which is enrolling urothelial patients. Normal prostate, bladder and pancreas expression is also an on-target risk that matters more than usual in a man with a prostate primary and prior pancreatitis. Test only if a PSCA protocol opens to this disease.

Recommended assay. PSCA IHC on tumor tissue (IHC · Archival block · 1-2 weeks)

A positive result would open.

- PSCA-directed CAR-T (City of Hope program) (CAR-T · trial only)
- BPX-601 (CAR-T · trial only)

Next step. Defer

SLITRK6 protein expression

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

SLITRK6 is expressed on urothelial carcinoma and was developed specifically in this disease, so it belongs on a urothelial gap list even though the program behind it stalled.

SLITRK6 is genuinely urothelial-restricted but the only binder that reached patients has not advanced past phase 1, so a positive stain would open nothing that is currently enrolling. With one block left and several higher-yield stains competing for it, this is the wrong slide to spend. Revisit if a SLITRK6 program reopens.

Recommended assay. SLITRK6 IHC on tumor tissue (IHC · Archival block · 1-2 weeks, and the assay is not routinely offered)

A positive result would open.

- sirtratumab vedotin (SLITRK6-directed ADC · trial only)

Next step. Defer

STEAP1 protein expression

Priority: low **Scope:** Relevant to this tumor type **Status:** Not measured

STEAP1 is expressed in bladder cancer alongside its main prostate indication, and the patient has a second prostate primary under surveillance. It has never been assessed.

STEAP1 programs are running in castration-resistant prostate cancer, and this patient's prostate primary has no evidence of disease, so neither his urothelial disease nor his prostate history gives him a route to one, and some normal urothelial expression complicates the tumor-versus-normal argument on top of that. There is nothing to order here today.

Recommended assay. STEAP1 IHC on tumor tissue (IHC · Archival block · 1-2 weeks)

A positive result would open.

- xaluritamig (STEAP1 x CD3 bispecific · trial only)

Next step. Defer

Measured, but not to decision resolution

A result exists for each of these, but not at the resolution a treatment decision needs. The workup that would close each gap is carried into the Target validation paths report.

HER2 protein overexpression (IHC 3+)

Priority: essential

On file. profile.json:biomarkers[0] HER2, confirmation_status 'ihc_pending'; profile.json:biomarkers[0] three concordant IHC 3+ results (primary 2023-11, liver met 2023-11, recurrence 2025-03), clone 4B5 and a national reference lab; profile.json:biomarkers[0] discordant weak HER2 on research-grade MXIF of 2026-08 nodal tissue; repeat validated IHC performed 2026-08, tier not in the record; profile.json:targetable_features[0] names HER2 reconciliation as the near-term decision gate

Gap. The 3+ calls describe 2023 and 2025 tumor. The current (2026-08) nodal tumor has been read only by research-grade multiplex immunofluorescence, which is not a validated HER2 assay and scored weak. Closing the gap needs the tier from the validated HER2 IHC drawn 2026-08 (4B5 or equivalent, reflex ISH if 2+), reported with tumor content stated, since the nodal cores carry only 5-10% tumor.

HER2 status on the current tumor is the open question, not HER2 status in 2023. The validated repeat IHC was already drawn in 2026-08 and its tier is simply not in the record, so the first move is to retrieve it and not to cut a new stain. Until that tier is in hand, continuing a poorly tolerated T-DXd rests on three-year-old tissue, and a genuine drop to HER2-low would move the drug off its label and onto a bystander-payload argument.

Tumor mutational burden, high (TMB-H)

Priority: high

On file. profile.json:biomarkers[9] TMB 12.4 mut/Mb in primary and liver metastasis, confirmation_status 'confirmed'; profile.json:biomarkers[9] method 'institutional hybrid-capture tumor/normal NGS panel (468-gene class), 2023-11'; profile.json:biomarkers[9] decision_resolution already flags the cross-assay calibration hedge

Gap. The value comes from an institutional 468-gene hybrid-capture panel. The tumor-agnostic pembrolizumab indication reads TMB-H by an FDA-approved test, and FoundationOne CDx is still the only approved TMB companion diagnostic as of 2026-08, so 12.4 mut/Mb on a laboratory-developed panel does not satisfy the label as written. TMB recomputed from the whole-exome sequencing already performed, or a companion-diagnostic-grade panel report, would settle it.

This case does carry a TMB-H value, which is easy to miss because it sits inside a 2023 report, but it was generated on an assay the tumor-agnostic label does not recognise. That distinction bites in two places: trials that gate on TMB-H by a specified assay, and the ipilimumab-addition strategy the team is already weighing, where the strength of the neoantigen argument depends on the number holding up. Recomputing TMB from the whole-exome data already generated costs no tissue.

Mesothelin (MSLN) protein expression

Priority: high

On file. profile.json:biomarkers[4] mesothelin strong on research-grade MXIF 2026-08, confirmation_status 'ihc_pending'; profile.json:biomarkers[4] decision_resolution asks for validated mesothelin IHC before trial screening; profile.json:targetable_features[3] treats this as a confirmation-first lead

Gap. The only evidence is research-grade multiplex immunofluorescence, not a validated clinical mesothelin IHC, and mesothelin is atypical in urothelial carcinoma, which raises the bar on assay specificity. A validated stain on tumor tissue is what mesothelin-directed trials will ask for, and the 5-10% tumor content of the nodal cores has to be stated on the report.

A strong mesothelin signal in urothelial carcinoma is unusual enough that it either opens a serious cell-therapy option or turns out to be an artefact of an unvalidated assay, and only a validated stain separates those. The patient is actively pursuing cell therapy and already has apheresis and PBMCs banked, so a confirmed result would be immediately usable for screening. Screening on the MXIF result alone risks consenting to a protocol the tumor cannot support.

PRAME expression (HLA-A*02:01-restricted)

Priority: high

On file. profile.json:biomarkers[7] PRAME-directed imaging completed 2026-08, result not in the available record, confirmation_status 'unknown'; profile.json:biomarkers[12] HLA-A*02:01 positive by high-resolution typing 2025-01, confirmation_status 'confirmed'; profile.json:targetable_features[5] describes this axis as conditional on expression

Gap. The 2026-08 PRAME imaging read is not in the record, and imaging is not what a PRAME trial screens on. Trials require expression by validated IHC or RT-PCR at a stated threshold, which has not been done. Retrieve the imaging read, then run the expression assay the candidate protocol names.

The expensive half of this question, the HLA type, is already answered, and PRAME expression is the only thing standing between this patient and a class of agents he is a plausible candidate for. PRAME in urothelial carcinoma is a subset finding rather than the rule, so a negative is a realistic and useful result. The MAGE-A4 experience is the template: HLA matched, expression negative, option closed by a test and not by a guess.

Nectin-4 (NECTIN4) protein expression

Priority: high

On file. profile.json:biomarkers[2] Nectin-4 95% membranous (primary 2023-11), 100% in the in-situ component (2025-03), confirmation_status 'confirmed'; profile.json:biomarkers[2] Nectin-4-directed imaging performed 2026-06, result not in the available record; profile.json:targetable_features[1] flags acquired Nectin-4 downregulation after enfortumab vedotin as plausible and unverified

Gap. Both stains predate 25 months of enfortumab vedotin. Antigen downregulation is a recognised route to EV resistance, so the historical 95-100% result cannot carry a re-targeting decision. Closing the gap means the 2026-06 Nectin-4 imaging read, or a contemporary IHC on post-EV tissue.

Nectin-4 was strongly positive before treatment and its current state is unverified, which is exactly the situation where a stale stain misleads. The 2026-06 imaging study has already been done and its read would answer the question without touching the scarce tissue. Retained expression keeps next-generation Nectin-4 agents with non-MMAE payloads on the table, while loss would close that axis cleanly instead of leaving it open as a standing maybe.

TROP2 (TACSTD2) protein expression

Priority: high

On file. profile.json:biomarkers[3] TROP2 low positive (weak) on research-grade MXIF 2026-08, confirmation_status 'ihc_pending'; profile.json:biomarkers[3] validated TROP2 IHC performed 2026-08 at a tertiary academic center, tier not in the available record

Gap. A weak MXIF signal is recorded as low expression and not as a negative, and multiplex immunofluorescence is not a validated TROP2 assay. The validated IHC drawn 2026-08 needs to be retrieved and reported as a tier, percent times intensity or H-score, so that trials gating on expression can be screened against a real number.

Weak TROP2 on a research assay is a low-positive, and a low-positive predicts ADC benefit less reliably than strong expression, so the number the validated stain gives changes how much weight this axis deserves. TROP2-ADC development in urothelial cancer has not been strictly expression-gated, which means low expression does not close the door, and the withdrawal of the sacituzumab govitecan urothelial indication means the realistic route here is a trial. The stain has already been performed; what is missing is the report.

MUC1 protein expression

Priority: medium

On file. profile.json:biomarkers[5] MUC1 strong on research-grade MXIF 2026-08, confirmation_status 'ihc_pending'; profile.json:biomarkers[5] decision_resolution asks for validated MUC1 IHC with clone and threshold per candidate trial

Gap. Research-grade multiplex immunofluorescence with no validated tier. MUC1 is expressed at low level on normal epithelia, so a strong signal only becomes meaningful when the assay distinguishes the tumor-associated glycoform the candidate agent targets. The stain has to be the clone that protocol names.

MUC1 sits behind mesothelin in this case: same research assay, same missing validation, thinner therapeutic bench. It is worth one slide on the block being cut anyway, mainly because a MUC1-directed cell therapy or engager trial would demand its own validated read at screening. Until then the strong signal is a lead and not a result.

PD-L1 protein expression

Priority: medium

On file. profile.json:biomarkers[1] CPS 5 and 5% tumor cells on the primary (2023-11), CPS 0 on the liver metastasis (2023-11), confirmation_status 'confirmed'; profile.json:biomarkers[1] method 'IHC E1L3N clone, validated comparable to 22C3 pharmDx'; profile.json:biomarkers[1] decision_resolution notes heterogeneity weakens PD-L1 as a predictor here

Gap. Two sites gave CPS 5 and CPS 0, which is discordance and not a number, and the stain used a laboratory clone validated as comparable to 22C3 pharmDx rather than the companion assay itself. No PD-L1 result exists on tissue from 2025 or later. A trial specifying CPS at or above 10 by 22C3 would need a fresh companion-assay read on contemporary tissue.

PD-L1 has been measured but the result is heterogeneous and dates from 2023, so it cannot support a CPS-gated decision as it stands. Its practical weight here is low: urothelial checkpoint use is not PD-L1 gated and the patient has already had both a deep response and progression on PD-1 blockade. Repeat the stain when a specific protocol asks for a specific clone and cutoff, not before.

Fibroblast activation protein alpha (FAP), stromal

Priority: low

On file. profile.json:biomarkers[6] FAP-positive stroma on research-grade MXIF 2026-08, confirmation_status 'unknown'; profile.json:biomarkers[6] FAPI PET 2025-09 showed intense uptake in prostate and seminal vesicle only, attributed to biopsy-proven BCG granulomatous inflammation, with no FAP-avid urothelial disease; profile.json:biomarkers[6] repeat FAP imaging 2026-06, result not in the available record

Gap. Stromal positivity on an unvalidated multiplex assay conflicts with a FAPI PET that showed no avid urothelial lesion, and the uptake it did show sits in granulomatous tissue, a classic false positive. What would settle it is quantitative FAPI PET uptake in the known nodal disease, read with the post-SBRT inflammatory caveat stated.

FAP here is an imaging-research lead, not a treatment lead: the one FAPI PET on record found no FAP-avid urothelial disease and lit up granulomatous prostate tissue instead. The 2026-06 repeat study is worth retrieving because it has already been done, but a stromal signal on multiplex immunofluorescence will not qualify anyone for a radioligand protocol. Recent SBRT to the nodal disease adds treatment-related fibroblast activation, which makes any new uptake harder to attribute.

Already established

- **Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)** — profile.json:biomarkers[10] MSI: microsatellite stable in both specimens, method 'institutional NGS panel (2023-11)', confirmation_status 'confirmed'; concordant with profile.json:biomarkers[9] TMB 12.4 mut/Mb, which is elevated but well below the hypermutated range dMMR produces
- **HLA class I genotype** — profile.json:biomarkers[12] HLA-A*02:01 positive, high-resolution class I and II typing 2025-01, confirmation_status 'confirmed'; profile.json:biomarkers[12] full typing withheld from the derived profile as quasi-identifying, which does not affect the A*02:01 call
- **MAGE-A4 expression (HLA-A*02-restricted)** — profile.json:biomarkers[8] MAGE-A4 tumor expression negative (2024-06), confirmation_status 'confirmed'; profile.json:biomarkers[8] testing was performed as part of screening for a MAGE-A4 TCR trial for which the patient was HLA-matched

Panel targets out of scope for this tumor

58 of 72 panel targets were screened and set aside as having no plausible connection to this patient's tumor type or stated features: ALPPL2 · ANTXR1 · AXL · CA9 · CD19 · CD200 · CD37 · CD46 · CD70 · CDCP1 · CDH17 · CDH3 · CDH6 · CEACAM5 · CLDN18 · CLEC12A · CT83 · DLK1 · DLL3 · EGFR · EGFR (variant III) · ENPP3 · EPCAM · EPHA2 · ERBB2 · ERBB3 · F3 · FOLH1 · FOLR1 · FZD7 · GPC3 · GPRC5D · GUCY2C · HAVCR1 · IL13RA2 · IL3RA · ITGB6 · L1CAM · LGR5 · LILRB4 · LY6E · LYPD3 · MAGEA1 · MAGEC2 · MERTK · MET · MMP14 · PODXL · PTK7 · ROR1 · SEZ6 · SLC34A2 · SLC44A4 · SSTR2 · SSX2 · TNFRSF10B · TNFRSF17 · TSPAN8